

DEPARTMENT OF COMPUTER & INFORMATION SYSTEMS ENGINEERING

BACHELORS IN COMPUTER SYSTEMS ENGINEERING

Course Code: CS-220

Course Title: Digital Logic Design

Complex Engineering Activity

SE Batch 2024, Fall Semester 2025

Grading Rubric

DESIGN PROJECT

Group Members:

Student No.	Name	Roll No.
S1	Syeda Maham Fatima	CS-24054
S2	Syeda Umaira Mashhood	CS-24055
S3	Nawal Abid	CS-24061

PLO 11 Project management – Rubric for Affective Domain Assessment (A-3)

CRITERIA AND SCALES					Marks Obtained		
Criterion 1: <u>Acknowledge</u> the importance of project management. [5 marks]							
1	2	3	4	5			
Doesn't acknowledge the importance of project management.	Rarely acknowledges the importance of project management.	Partially acknowledges the importance of project management.	Fairly acknowledges the importance of project management.	Fully acknowledges the importance of project management.			
Criterion 2: <u>Practice</u> concepts and principles of project management. [5 marks]							
1	2	3	4	5			
Doesn't practice concepts and principles of project management.	Rarely practices concepts and principles of project management.	Partially practices concepts and principles of project management.	Fairly practices concepts and principles of project management.	Fully practices concepts and principles of project management.			
Criterion 3: <u>Value</u> the need for project management. [5 marks]							
1	2	3	4	5			
Doesn't value the need for project management.	Rarely values the need for project management.	Partially values the need for project management.	Fairly values the need for project management.	Fully values the need for project management.			
Psychomotor Domain Assessment Rubric-Level P3							
CRITERIA AND SCALES					Marks Obtained		
Criterion 1: <u>Equipment Identification</u> [5 marks]							
1	2	3	4	5			
Not able to identify the equipment.	--	--	--	Able to identify equipment as well as its components.			
Criterion 2: <u>Equipment Use</u> . [5 marks]							
1	2	3	4	5			
Doesn't demonstrate the use of equipment.	Slightly demonstrates the use of equipment.	Somewhat demonstrates the use of equipment.	Moderately demonstrates the use of equipment.	Fully demonstrates the use of equipment.			
Criterion 3: <u>Procedural Skills</u> . [5 marks]							
1	2	3	4	5			

Not able to either learn or perform lab work procedure.	Able to slightly understand lab work procedure and perform lab work.	Able to somewhat understand lab work procedure and perform lab work.	Able to moderately understand lab work procedure and perform lab work.	Able to fully understand lab work procedure and perform lab work.			
Criterion 4: Response. [5 marks]							
1	2	3	4	5			
Not able to imitate the lab work.	Able to slightly imitate the lab work.	Able to somewhat imitate the lab work.	Able to moderately imitate the lab work.	Able to fully imitate the lab work.			
Criterion 5: Observation's Use. [5 marks]							
1	2	3	4	5			
Not able to use the observations from lab work for experimental verifications and illustrations.	Slightly able to use the observations from lab work for experimental verifications and illustrations.	Somewhat able to use the observations from lab work for experimental verifications and illustrations.	Moderately able to use the observations from lab work for experimental verifications and illustrations.	Fully able to use the observations from lab work for experimental verifications and illustrations.			
Criterion 6: Safety Adherence. [5 marks]							
1	2	3	4	5			
Doesn't adhere to safety procedures.	Slightly adheres to safety procedures.	Somewhat adheres to safety procedures.	Moderately adheres to safety procedures.	Fully adheres to safety procedures.			
Criterion 7: Equipment Handling. [5 marks]							
1	2	3	4	5			
Doesn't handle equipment with required care.	Rarely handles equipment with required care.	Occasionally handles equipment with required care.	Often handles equipment with required care.	Handles equipment with required care.			
Criterion 8: Group Work. [5 marks]							
1	2	3	4	5			
Doesn't participate and contribute.	Slightly participates and contributes.	Somewhat participates and contributes.	Moderately participates and contributes.	Fully participates and contributes.			

➤ Group Members:

Student No.	Name	Roll No.
S1(Group Leader)	Syeda Maham Fatima	CS-24054
S2	Syeda Umaira Mashhood	CS-24055
S3	Nawal Abid	CS-24061

➤ Project Title:

Finite State Machine (FSM) Sequence Detector

➤ Objective:

The objective of this project is to design and implement a synchronous **Mealy Finite State Machine (FSM)** that detects the binary sequences **001** and **010** from a serial input bit stream. The FSM outputs logic '1' at the exact moment either sequence is completed, since the Mealy design produces output based on both the **current state** and **current input**.

The detector is implemented using D flip-flops and combinational logic to achieve correct sequence recognition and proper state transitions.

➤ Detected Sequences:

Sequence 1: 001

Sequence 2: 010

These two sequences are 3 bits long as required and demonstrate overlapping, which makes the design more realistic and meaningful.

➤ Basic Features:

a) Mealy Machine Design:

The FSM is a **Mealy type detector**, meaning the output depends on the current state **and** the incoming bit. This makes detection faster because the output becomes '1' immediately when the final bit of the sequence is received.

b) Immediate Sequence Detection:

The output turns HIGH in the same clock cycle when the last bit of 001 or 010 arrives, making the system efficient and responsive.

c) Overlapping Sequence Handling:

The FSM is designed to correctly handle overlapping sequences in a continuous bit stream.
Example: A bit stream like **0010** contains **both** sequences, and the FSM detects them consecutively without missing any.

d) Synchronous Operation with Flip-Flops:

State transitions occur on every rising edge of the clock, ensuring stable and predictable operation. D Flip Flops (7474) are used to store state bits.

e) Hardware Implementation:

The circuit is implemented using:

1. 7474 IC which is D flip-flop
 2. AND / XOR / NOT gates
 3. Input switches
 4. Output indicators (LED or seven-segment)
- The design is tested and verified using Electronics Workbench

f) Clear State Representation:

A well defined state diagram and state transition table guide the circuit design and make the system easier to understand and troubleshoot.

➤ Optional / Additional Features:

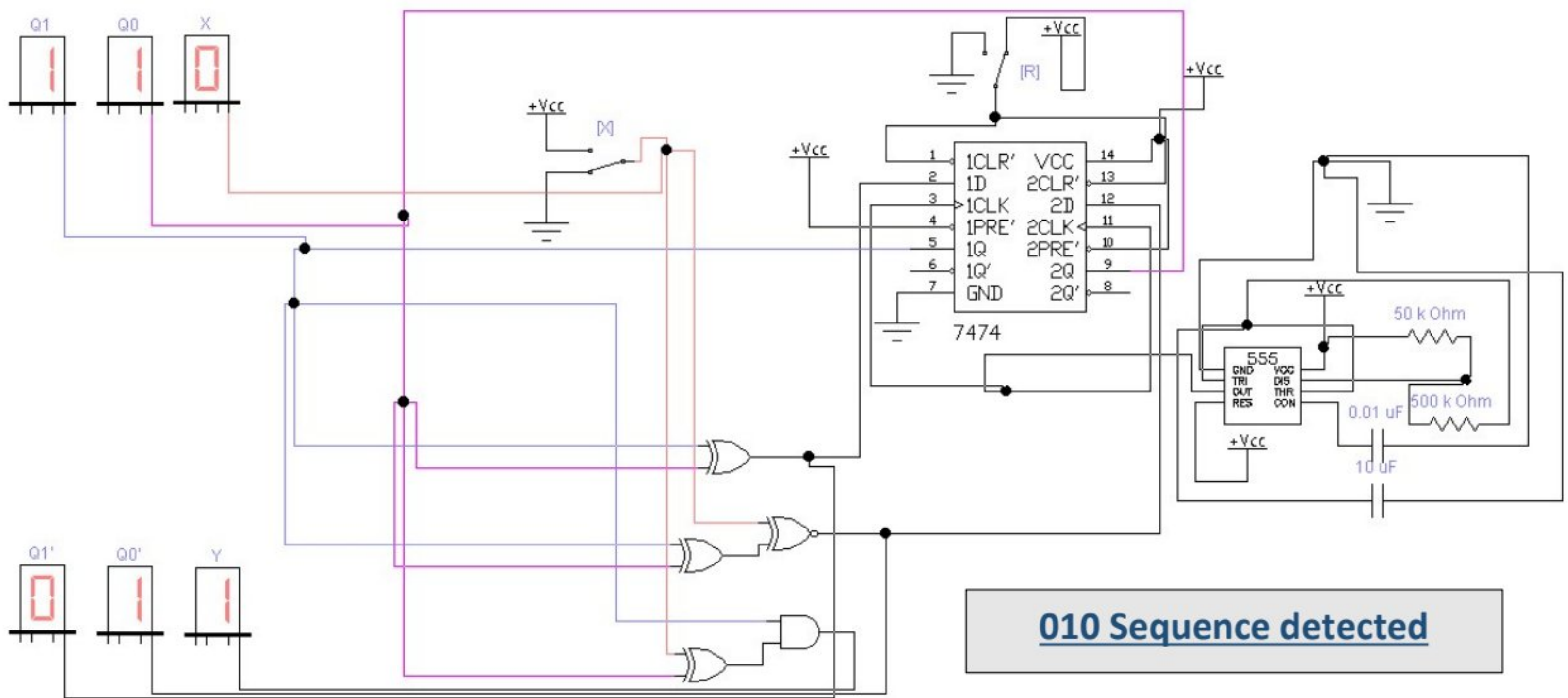
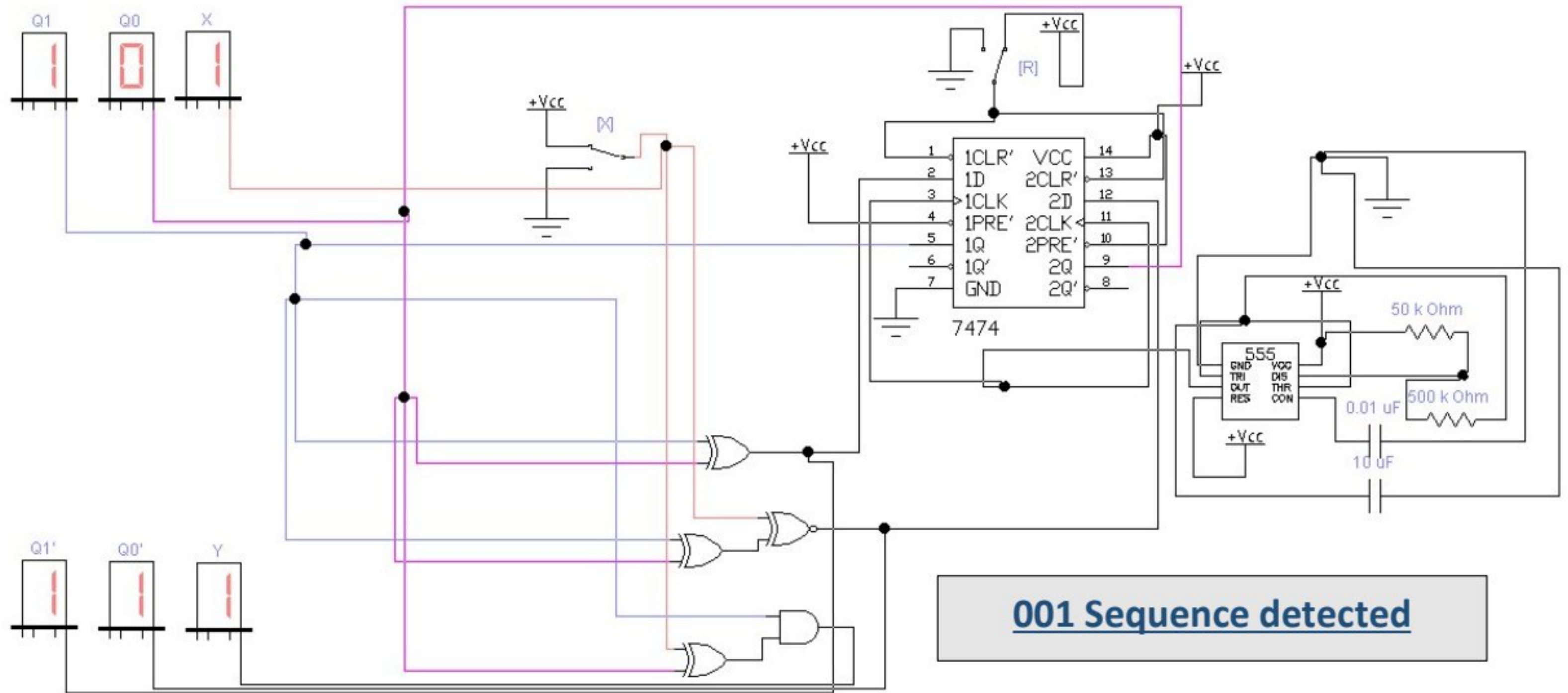
1. Seven-Segment Display Output:

A seven segment display is added to visually show the current state or detection output. This improves readability and makes demonstrations more attractive and interactive.

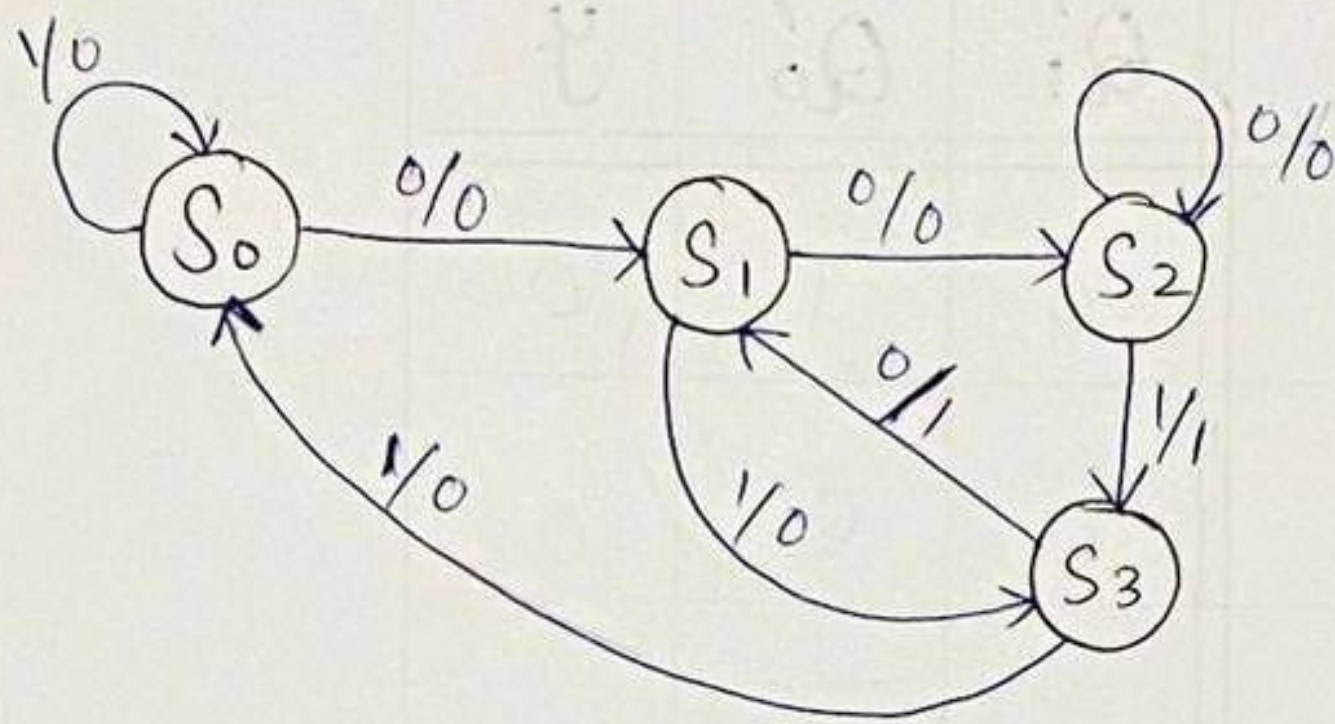
2. Clock Generation using 555 Timer (Optional):

A 555 Timer IC can be used in **Astable Mode** to automatically generate clock pulses for the FSM. This removes the need for manual clocking and makes the circuit fully autonomous during operation.

TESTING USING ELECTRONICS WORKBENCH



STATE TRANSITION DIAGRAM:-



SEQUENCE TO
DETECT:
0 0 1
0 1 0

where S_0, S_1, S_2, S_3 are states and
0/0, 0/1, 1/0, 1/1 indicates Input/Output

STATE TABLE:-

		x	
Q_1	Q_0	0	1
0	0	01, 0	00, 0
0	1	10, 0	11, 0
1	0	10, 0	11, 1
1	1	01, 1	00, 0

STATE ASSIGNMENT:-

	Q_1	Q_0
S_0	0	0
S_1	0	1
S_2	1	0
S_3	1	1

Q_1	Q_0	x	Q_1'	Q_0'	y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	0	1	1
1	1	1	0	0	0

Here;

Q_1 = Current State Variable

Q_0 = Current State Variable

x = Input

Q_1' = Next State Variable

Q_0' = Next State Variable

y = Output

IMPLEMENTATION AND

CIRCUIT DIAGRAM:-

Since D flip flops are used, so the output will be same as the input.

$$D_1 = Q_1' \quad \text{and} \quad D_0 = Q_0'$$

EQUATION FOR D_1 : (Using K-Map for 3 Variables)

D_1

$Q_1 \backslash Q_0 x$	00	01	11	10
0	0	1	1	1
1	1	1	0	0

$$D_1 = \bar{Q}_1 Q_0 + Q_1 \bar{Q}_0$$

$$D_1 = Q_1 \oplus Q_0$$

EQUATION FOR D_0 : (Using K-Map for 3 Variables)

D_0

$Q_1 \backslash Q_0 x$	00	01	11	10
0	1	0	1	0
1	0	1	0	1

$$\begin{aligned} D_0 &= \bar{Q}_1 \bar{Q}_0 x + \bar{Q}_1 Q_0 x + Q_1 \bar{Q}_0 x + Q_1 Q_0 x \\ D_0 &= \bar{x} (\bar{Q}_1 \bar{Q}_0 + Q_1 Q_0) + x (\bar{Q}_1 Q_0 + Q_1 \bar{Q}_0) \\ D_0 &= \bar{x} (Q_1 \odot Q_0) + x (Q_1 \oplus Q_0) \\ D_0 &= \bar{x} (\overline{Q_1 \oplus Q_0}) + x (Q_1 \oplus Q_0) \end{aligned}$$

$$D_0 = x \odot (Q_1 \oplus Q_0)$$

EQUATION FOR y : (Using K-Map for 3 Variables)

y

$Q_1 \backslash Q_0 x$	00	01	11	10
0	0	0	0	0
1	0	1	0	1

$$\begin{aligned} y &= Q_1 \bar{Q}_0 x + Q_1 Q_0 \bar{x} \\ y &= Q_1 (\bar{Q}_0 x + Q_0 \bar{x}) \end{aligned}$$

$$y = Q_1 (Q_0 \oplus x)$$

IMPLEMENTATION AND

Q_1 Q_0 x

